

GDOR-SLAM: Guarded Dynamic Observation Recovery for RGB-D Gaussian SLAM

Author List To Be Finalized

Abstract—Dynamic masks protect a persistent Gaussian map, but hard exclusion can also remove background and camera-consistent observations that remain useful for localization. This paper studies a task mismatch in dynamic RGB-D Gaussian SLAM: an observation that is useful for transient camera tracking is not necessarily eligible to modify a static world model. We present GDOR-SLAM, which implements separate tracking-support and persistent-mapping routes for dynamic-region evidence. Temporal depth consistency, residual-flow consistency, and tracking-risk state qualify a conservative subset of excluded observations for current-frame feature extraction and pose estimation, while the original conservative mask continues to govern ORB landmark insertion and Gaussian initialization, optimization, and densification. The method therefore recovers tracking support without granting recovered dynamic-region observations persistent map authority. Evaluation includes a 63-run TUM/Bonn same-source protocol, a separately frozen 36-cell TUM4 validation, a 9-run map-matched no-track-reuse control, and a single-seed common-view mapping diagnostic. GDOR reduces the 63-run All7 mean ATE from 11.331 cm to 3.798 cm relative to semantic hard exclusion. The sequence-median ATE shifts from 4.278 cm to 3.261 cm, GDOR has a 7/0/0 win/tie/loss record over Semantic, and the largest failure-case rescue is Bonn crowd2 at 52.184 cm to 3.226 cm. On the frozen TUM4 protocol, GDOR reduces the grand mean from 9.746 cm to 3.415 cm; the paired-cell median improvement is 0.209 cm with a 9/0/3 win/tie/loss record, driven mainly by the sitting_halfsphere failure case at 32.374 cm to 7.354 cm. The matched control uses the same persistent mapping weight but disables qualified tracking recovery; it obtains 7.816 cm mean ATE, compared with 4.053 cm for GDOR on the tested subset, with 0.864 cm median sequence improvement and a 3/0/0 win/tie/loss record. These results support the concrete guarded recovery and two-route mechanism, while conceptual priority, protocol-matched superiority over official DyPho-SLAM, and multi-seed mapping superiority are not claimed.

Index Terms—3D Gaussian splatting, dynamic RGB-D SLAM, visual tracking, dynamic observation recovery, Gaussian mapping.

I. INTRODUCTION

3D Gaussian Splatting (3DGS) provides an explicit differentiable representation for dense scene rendering and has become a practical substrate for online visual world modeling [1]. Gaussian SLAM systems such as Photo-SLAM, SplatAM, GS-SLAM, and Gaussian-SLAM combine camera tracking with dense Gaussian reconstruction [2]–[5]. Their tracking frontends and persistent maps, however, still depend on a predominantly static world. When people or objects occupy a large image region, dynamic observations can corrupt camera estimation and produce persistent Gaussian artifacts.

Semantic and geometric exclusion is therefore a natural defense. Classical dynamic SLAM removes features associated with movable classes [6]–[8], while recent dynamic Gaussian

SLAM systems combine semantics, temporal consistency, depth warping, photometric evidence, or uncertainty to suppress unsafe observations [9]–[12]. This policy is appropriate for static map construction, but it can be too restrictive for tracking. A semantic mask can cover visible background, object boundaries, temporarily stationary surfaces, or pixels whose image motion is dominated by camera motion. Removing all such observations can leave the pose estimator poorly constrained exactly when dynamic occlusion is strongest.

The key issue is not only whether an observation is static, but what authority it should receive. Tracking and mapping have different failure costs and persistence. A locally consistent observation may be useful for estimating the current camera pose while still being too uncertain to initialize or optimize persistent static structure. Treating both decisions with one binary mask conflates *tracking utility* with *map eligibility*.

We propose **GDOR-SLAM**, a dynamic RGB-D Gaussian SLAM system based on Guarded Dynamic Observation Recovery (GDOR). GDOR evaluates temporal depth consistency, residual optical-flow consistency, and the current tracking-risk state to recover a conservative subset of observations inside excluded dynamic regions. Recovered observations may replenish features and support the current camera estimate. They do not modify the conservative mask used for persistent ORB landmarks or Gaussian initialization, update, and densification. The method therefore separates transient tracking privilege from persistent map privilege.

Our contributions are:

- We instantiate an auditable two-channel observation contract for RGB-D/ORB Gaussian SLAM: tracking utility does not automatically grant persistent Gaussian map authority.
- We introduce GDOR, which combines temporal depth, residual-flow, and tracking-risk evidence to selectively recover dynamic-region observations for transient tracking.
- We route the recovered tracking mask and conservative mapping weight independently; the release implements explicit recovered-support overlap, MapPoint-rejection, and RGB-D densification-rejection audits.
- We provide evidence from a 63-run same-source TUM/Bonn protocol, a frozen 36-cell TUM4 validation, a 9-run map-matched causal control, and a common-view static-region mapping diagnostic.

The supported claim is deliberately bounded. GDOR improves the same-source semantic parent in the declared tracking protocols and remains better than a matched persistent-weight/no-track-reuse control on three tested sequences. The current assets do not constitute an official DyPho-SLAM rerun,

GDOR-SLAM: guarded tracking support without persistent dynamic writes

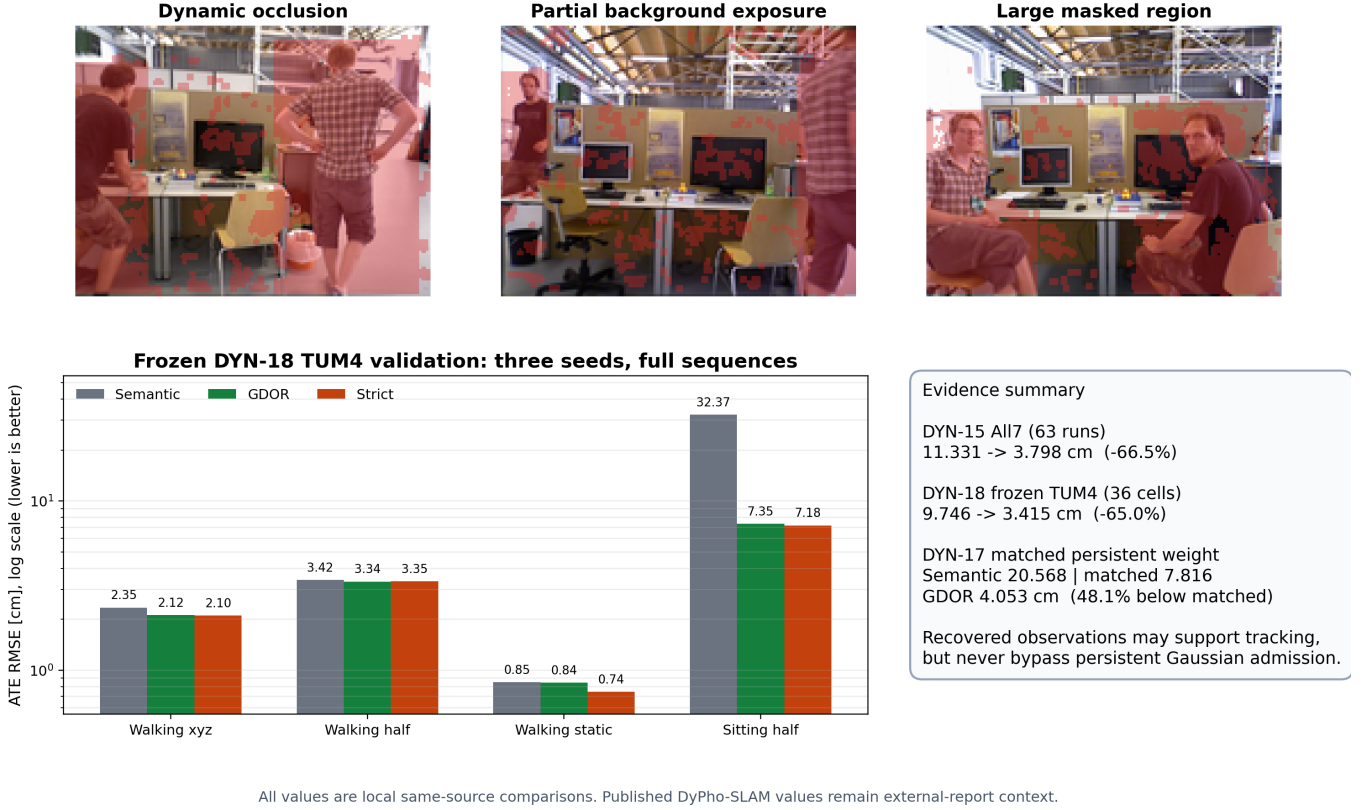


Fig. 1. GDOR-SLAM overview and evidence summary. Real TUM dynamic frames show large semantic exclusion regions. The frozen TUM4 validation improves all four sequence means over the same-source Semantic parent, while the 63-run and map-matched control summaries show the broader tracking effect and the role of tracking-side recovery. Published DyPho-SLAM values remain external-report context.

and the mapping evaluation is single-seed. The paper therefore argues for a concrete RGB-D/ORB recovery and routing mechanism rather than conceptual priority for tracking-mapping separation, universal improvement, or external SOTA superiority.

II. RELATED WORK

A. Visual SLAM in Dynamic Environments

Classical feature and dense SLAM systems rely on robust estimation but still assume that most observations belong to a static scene [13]–[16]. Semantic dynamic SLAM makes this assumption explicit by removing features or pixels associated with potentially moving classes. DynaSLAM combines learned instance masks with multi-view geometry [6], while DS-SLAM and Dynamic-SLAM use semantic predictions to reject dynamic correspondences [7], [8]. Residual-based reconstruction instead detects observations that disagree with the estimated static model, as in ReFusion [17].

Motion-aware systems retain more information about the moving scene. VDO-SLAM estimates camera and object motion from dynamic observations [18], and optical-flow tracking provides a standard source of short temporal correspondences [19]. The central difficulty is identifiability: image motion alone mixes camera motion, object motion,

and correspondence error. Our method uses camera-motion inconsistency as a guarded cue, but the main distinction is role assignment: recovered dynamic-region observations may support short-term tracking while remaining ineligible for persistent static-map landmarks.

B. Dense Neural and Gaussian SLAM

Dense neural SLAM has represented scene geometry with implicit fields or feature grids [20]–[22], while 3DGS provides an explicit representation with differentiable rasterization [1]. Photo-SLAM couples ORB-SLAM3 with a Gaussian mapper [2]; SplaTAM jointly tracks and maps RGB-D observations with Gaussians [3]; GS-SLAM and Gaussian-SLAM develop online Gaussian densification and optimization strategies [4], [5]. These methods establish dense Gaussian mapping in static scenes but do not by themselves resolve whether a measurement should influence camera tracking, map optimization, both, or neither in a dynamic scene.

C. Dynamic Gaussian SLAM

The closest static-output Gaussian SLAM systems suppress dynamic evidence using combinations of semantics, temporal consistency, and geometry. DG-SLAM fuses semantic and multi-frame depth-warp masks and refines a coarse pose

against the Gaussian map [9]. DGS-SLAM refines segmentation through photometric consistency and filters dynamic content during Gaussian insertion and keyframe selection [10]. DyGS-SLAM improves dynamic-scene localization and Gaussian reconstruction in real time [23], and MPDG-SLAM introduces motion-probability reasoning for dynamic 3DGS-SLAM [24]. DyPho-SLAM uses refined masks and adaptive feature extraction for real-time dynamic Gaussian SLAM [11]. DAGS-SLAM maintains a spatiotemporal motion probability for each Gaussian and schedules semantic inference from uncertainty [12]. Concurrent DL-SLAM explicitly uses transiently static objects for pose estimation while suppressing their persistent contribution through dual-level probabilities [25]. This establishes tracking–mapping role separation as a shared recent direction rather than a conceptual priority claim of GDOR.

Dynamic-output systems address a different objective. D²GSLAM combines static 3D Gaussians with dynamic 4D Gaussians and uses motion information for progressive pose refinement [26]. Our output remains a static Gaussian map. Rather than reconstructing object trajectories, estimating object-level probabilities, or injecting object-motion priors, GDOR instantiates a narrower RGB-D/ORB mechanism: binary local recovery from temporal depth and residual-flow checks, separate `staticMask/staticWeight` routes, hard persistent-weight gating for MapPoints and Gaussians, and an explicit recovered-support overlap audit. The contribution is this auditable implementation and its matched no-track-reuse control, not the first statement that tracking and mapping can use different dynamic evidence.

III. METHOD

A. Observation Routing Contract

GDOR-SLAM extends the RGB-D path of Photo-SLAM [2], with an ORB-SLAM3 frontend [14] and a 3DGS mapping backend [1]. All masks below are binary and defined per pixel $\mathbf{x}$. Let $M_t^{\text{sem}} = 1$ denote semantic exclusion and $M_t^{\text{flow}} = 1$ denote residual-motion exclusion. Their union and its morphological closing are

$$D_t^{\text{raw}} = M_t^{\text{sem}} \vee M_t^{\text{flow}}, \quad \bar{D}_t = \text{Close}(D_t^{\text{raw}}). \quad (1)$$

The claim-bearing configuration routes two different signals: a binary support mask S_t^{track} to ORB extraction and current-frame tracking, and a float weight W_t^{map} to persistent ORB/Gaussian mapping. The conservative mapping route is fixed as

$$W_t^{\text{map}}(\mathbf{x}) = 1 - \bar{D}_t(\mathbf{x}). \quad (2)$$

Thus, a tracking decision can recover support without clearing the persistent mapping exclusion.

B. Residual-Flow and Temporal-Depth Evidence

A semantic prior marks classes that may move. Forward–backward Lucas–Kanade tracks [19] are sampled on an image grid; tracks with more than 1.5 pixels of round-trip error are discarded. A partial affine model is fitted by RANSAC on semantic-static support, while a fundamental-matrix check

requires at least six model correspondences and at least 60% epipolar inliers. If either model is invalid, flow-based recovery fails closed. Otherwise, residuals unexplained by the affine model form M_t^{flow} , and the full-frame residual mask plus a safety neighborhood forms the recovery veto G_t .

Let z_t , z_t^r , and b_{t-1} be observed, rendered, and historical background depths. For valid inputs, the temporal prior is

$$b_t(\mathbf{x}) = \frac{w_h I_h b_{t-1}(\mathbf{x}) + w_r I_r z_t^r(\mathbf{x})}{w_h I_h + w_r I_r}, \quad (3)$$

where I_h and I_r indicate valid historical and rendered depth. Depth consistency is accepted when rendered support is valid and at least K valid observations in a local neighborhood $\mathcal{N}(\mathbf{x})$ satisfy

$$\mathcal{C}_t^{\text{depth}}(\mathbf{x}) = \left[\sum_{\mathbf{y} \in \mathcal{N}(\mathbf{x})} \mathbf{1}(|z_t(\mathbf{y}) - b_t(\mathbf{x})| < \tau_d) \geq K \right]. \quad (4)$$

C. Guarded Support Qualification

A raw-excluded pixel is eligible for tracking recovery only when every required signal is valid:

$$R_t^{\text{elig}}(\mathbf{x}) = D_t^{\text{raw}}(\mathbf{x}) \wedge \mathcal{C}_t^{\text{depth}}(\mathbf{x}) \wedge \neg G_t(\mathbf{x}). \quad (5)$$

After the same morphological refinement used by the frontend, the tracking support is

$$S_t^{\text{track}}(\mathbf{x}) = \neg \text{Close}(D_t^{\text{raw}} \wedge \neg R_t^{\text{elig}})(\mathbf{x}), \quad (6)$$

and the support actually recovered after refinement is $R_t = D_t^{\text{raw}} \wedge S_t^{\text{track}}$. Invalid depth, insufficient rendered support, or an invalid motion model sets $R_t^{\text{elig}} = 0$ and returns the conservative raw-mask behavior.

D. Risk-Conditioned Tracking Recovery

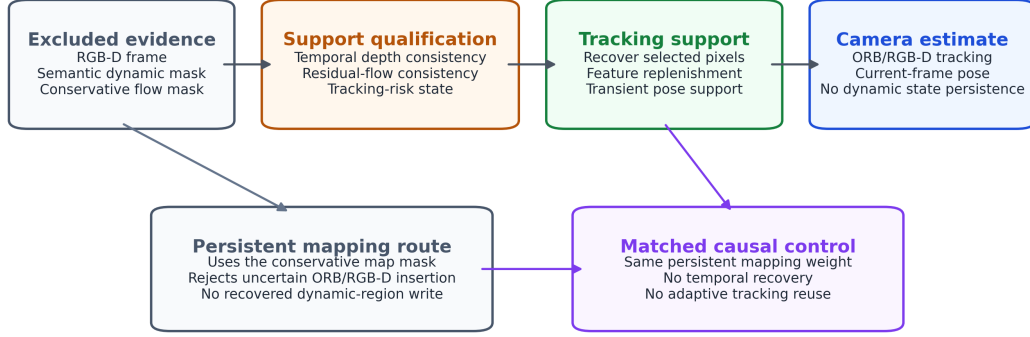
Recovery is useful only when the tracker needs additional support. GDOR therefore conditions feature replenishment on the previous optimized inlier count. During a tracking-risk window, the FAST threshold is temporarily relaxed and qualified pixels can participate in feature extraction and current-frame pose estimation. The relaxation persists for a short fixed hold interval to avoid frame-to-frame switching. Outside these windows, the system remains close to semantic hard exclusion.

The default Guarded configuration uses residual-flow protection, temporal-depth recovery, and risk-conditioned feature replenishment. The Strict configuration adds an RGB-D camera-guided flow model and tighter model-quality gates. Strict is retained as a stronger-intervention control, not as the proposed default, because stronger recovery is not uniformly more stable in the broader TUM/Bonn protocol.

E. Tracking–Mapping Decoupling and Matched Control

Recovered observations are transient. They may support current-frame ORB extraction and pose optimization, but the mapping route continues to use Eq. (2) for Gaussian loss weighting, RGB-D densification, and MapPoint-to-Gaussian admission. The route invariant is frame-local and admission-local: a frame that contains recovered tracking support does

GDOR-SLAM: guarded recovery with separate tracking support and mapping weight



Core contract: tracking support does not automatically grant persistent Gaussian map admission.

Fig. 2. GDOR-SLAM separates support qualification, transient tracking support, and persistent mapping weight. The matched no-track-reuse control uses the same persistent weight as GDOR but disables temporal recovery and adaptive tracking reuse.

not receive a persistent-map admission privilege. Every ORB MapPoint, Gaussian initialization, RGB-D densification candidate, and Gaussian loss term is still admitted or weighted only through W_t^{map} ; recovered support cannot set that weight to one, bypass keyframe static-weight checks, or waive the conservative mapping boundary. The implemented routing cases are summarized compactly as

configuration	S_t^{track}	W_t^{map}	
Semantic	$1 - \text{Close}(M_t^{\text{sem}})$	1 (default)	(7)
Matched	$1 - \bar{D}_t$	$1 - \bar{D}_t$	
GDOR/Strict	Eq. (6)	$1 - \bar{D}_t$	

The Semantic row has no supplied static weight, so the mapper reports its default unit support; this does not mean that the semantic tracking mask contains no exclusions.

The release audits the exact overlap

$$L_t = R_t \wedge \mathbf{1}[W_t^{\text{map}} \geq 0.5]. \quad (8)$$

For the claim-bearing routing, the intended invariant is $|L_t| = 0$. The counter is diagnostic and does not modify either route. Persistent ORB MapPoints are additionally checked against keyframe static weights, while Gaussian training and RGB-D densification consume the same bound weight.

To isolate tracking-side recovery from the raw-mask route, the map-matched no-track-reuse control uses the same $W_t^{\text{map}} = 1 - \bar{D}_t$ as GDOR, uses $1 - \bar{D}_t$ directly for tracking, and disables temporal recovery and adaptive feature replenishment. The GDOR–Matched comparison therefore tests qualified tracking reuse on top of the same persistent mapping weight; it is not a comparison between two Gaussian-admission masks.

F. Implementation Details

The frozen Guarded configuration applies semantic inference on every frame after a 50-frame system warmup and starts temporal background refinement after five frames. Lucas–Kanade tracking uses a 21×21 window, three pyramid levels,

and an 8-pixel sampling grid. The dominant-flow residual threshold is 1.5 pixels. Temporal depth combines rendered and historical background estimates with weights 0.6 and 0.4; accepted static observations update the stored background with weight 0.2. The local consistency threshold is 0.20 m. The local test uses a radius of two pixels, requires nine consistent neighbors, a minimum valid fraction of 0.02, and rendered opacity of at least 0.5. The prediction is disabled when the inter-frame motion exceeds 0.5 m translation or 0.7854 rad rotation.

Feature replenishment is activated when the previous optimized inlier count falls below 150. The FAST threshold is relaxed by a factor of 0.9 and the state is held for five frames. The flow guard uses a one-pixel conservative safety radius. All positive experiments keep object-motion pose priors disabled, use hard mapping masks, and prohibit soft dynamic Gaussian admission. These values were frozen before DYN-18 and were not adjusted using its outcomes. The added per-frame work is sparse LK/model fitting plus linear-time mask and depth tests; GDOR introduces no object state, pose prior, or additional persistent map variable.

IV. EXPERIMENTS

A. Protocols and Metrics

We evaluate GDOR-SLAM on dynamic RGB-D sequences from TUM RGB-D [27] and Bonn RGB-D Dynamic [17]. Four evidence blocks are kept separate.

DYN-15 is the primary same-source tracking protocol: Semantic, GDOR, and Strict are run with three seeds on TUM walking_xyz and all six Bonn dynamic sequences, for 63 completed runs. **DYN-18** is a separately sealed validation using the unchanged configurations on the four TUM sequences named in the DyPho-SLAM tracking table: walking_xyz, walking_halfsphere, walking_static, and sitting_halfsphere. It contains 36 planned cells after one exact Semantic sitting seed-0 repair; the initial failure and repair are retained in the audit.

DYN-17 evaluates the map-matched no-track-reuse control on three sequences and three seeds, for 9 completed runs. **DYN-16** is a seed-0 common-view mapping diagnostic on TUM walking_xyz and Bonn crowd, with 20 shared non-keyframe views per sequence.

All tracking runs process the full association file and retain invalid intervals in the all-input-frame trajectory. We report SE(3)-aligned ATE RMSE, one-frame translational RPE RMSE, failure rate, and end-to-end wall time. The mapping diagnostic evaluates PSNR, SSIM, and LPIPS on a fixed static-region mask shared by all compared configurations.

The benchmark harness stores the source snapshot, executable and library hashes, detector-engine hash, mask-configuration hash, association and ground-truth hashes, sequence, seed, command line, and output identity. The archived campaigns contain per-frame tracking state, recovery, insertion-gate, trajectory, runtime, and manifest records. The public release additionally contains the explicit overlap audit in Eq. (8); the archived DYN-15–18 summaries predate that field and therefore do not provide an empirical zero-overlap count. The DYN-18 protocol was sealed before evaluation; the one failed Semantic cell was repeated once without code or configuration changes in a fresh output root. The failure and repaired result both remain registered.

B. Tracking Accuracy and Frozen Generalization

Table I shows two complementary results. In DYN-15, GDOR reduces the All7 mean ATE from 11.331 cm to 3.798 cm, shifts the sequence-median ATE from 4.278 cm to 3.261 cm, and wins every declared sequence mean relative to Semantic (7/0/0 wins/ties/losses). The largest rescue is Bonn crowd2, where the mean falls from 52.184 cm to 3.226 cm. Across Bonn6, GDOR reduces the mean ATE by 68.2%. Because the mean is dominated by crowd2, Table I reports the failure case and median next to the mean rather than using the average alone.

DYN-18 tests the frozen method on a separate TUM4 campaign. GDOR improves all four sequence means over the same-source Semantic parent and wins 9 of 12 paired sequence/seed cells (9/0/3 wins/ties/losses). The median paired improvement is 0.209 cm; the corresponding two-sided exact sign test is not significant ($p = 0.146$). The improvement is small on walking_static and walking_halfsphere, but large on sitting_halfsphere, where Semantic suffers repeated tracking failures. Sitting_halfsphere contributes a 25.020 cm mean paired improvement and three wins, whereas the other nine cells yield six wins, a 0.102 cm mean improvement, and a 0.188 cm median improvement. We therefore interpret DYN-18 as conditional rescue of a failure regime plus near-identity behavior elsewhere, not as a uniformly large effect. Strict is slightly lower than GDOR on three DYN-18 rows, but it is not promoted to the main method because the broader DYN-15 protocol shows substantially worse mean ATE and higher runtime.

C. Matched Map-Admission Control

Table II addresses a central alternative explanation. Matched applies the raw semantic-plus-flow support to both tracking

and the persistent mapping weight, while GDOR keeps that mapping weight fixed and recovers only the qualified tracking support. Across the three tested sequences, the matched control obtains 7.816 cm mean ATE, compared with 4.053 cm for GDOR. The median sequence improvement is 0.864 cm, GDOR wins all three sequence means (3/0/0 wins/ties/losses), and the failure-case contrast is Bonn crowd2 at 13.597 cm for Matched versus 3.226 cm for GDOR. Thus, a fixed raw-mask tracking route with the same persistent mapping weight does not explain the reported GDOR result on this subset. The control does not separate temporal recovery from adaptive replenishment, but it isolates their combined tracking-side effect from a persistent mapping-weight change.

D. Safety, Efficiency, and Mapping Diagnostic

GDOR lowers the All7 failure rate but increases wall time from 32.46 s/run to 49.84 s/run. The overhead comes from residual-flow estimation, temporal-depth maintenance, and feature bookkeeping. Strict is slower and less stable in the broader protocol, which motivates the Guarded configuration as the accuracy–cost compromise.

Map support is not a quality score. It is the mean fraction of processed-frame pixels satisfying $W_t^{\text{map}} \geq 0.5$. Semantic supplies no static weight, so 100% denotes the mapper’s default unit weight rather than the static fraction of its tracking mask. For GDOR and Strict, the value records the conservative persistent mapping boundary; recovered tracking observations do not alter that weight. In the same DYN-15 audit, GDOR rejects 11.25% of considered ORB insertions and 11.94% of RGB-D Gaussian densification candidates at the static-mask boundary. These rejection counters show that the mapping gates are active, but they are not substitutes for the overlap counter in Eq. (8), which is unavailable in the archived summaries.

The DYN-16 common-view audit is a limited preservation diagnostic that checks whether this tracking intervention damages the declared static-region reconstruction views. GDOR improves the two-sequence online mean from 19.068 to 20.154 dB PSNR, from 0.690 to 0.740 SSIM, and from 0.354 to 0.287 LPIPS. Strict is marginally higher in PSNR. Because this audit uses one seed, two scenes, fixed static masks, and no independent ghost/completeness truth, it supports only a limited preservation diagnostic, not zero ghost, zero contamination, or a general mapping-superiority claim.

E. External Positioning and Limitations

Published DGS-SLAM, DG-SLAM, and DyPho-SLAM values are retained as related-work context, not mixed with the local tables as protocol-matched reruns. The DyPho-SLAM paper reports 1.6, 2.6, 0.6, and 1.6 cm on the four corresponding TUM sequences. GDOR is not below those external values in DYN-18. Differences in implementation, masks, preprocessing, hardware, and evaluation prevent a valid direct ranking. An official protocol-matched DyPho-SLAM rerun is therefore required before claiming external superiority.

The second limitation is mapping breadth. The current common-view evaluation is stricter than method-specific render selection because views and masks are shared, but it is

TABLE I

SAME-SOURCE TRACKING EVIDENCE. VALUES ARE THREE-SEED ATE RMSE IN CENTIMETERS; LOWER IS BETTER. DYN-15 IS THE 63-RUN TUM/BONN PROTOCOL. DYN-18 IS A SEPARATELY SEALED 36-CELL TUM4 VALIDATION. REDUCTION IS GDOR RELATIVE TO SEMANTIC. AGGREGATE ROWS INCLUDE THE REQUIRED DISTRIBUTIONAL COMPANION: FAILURE-CASE EXAMPLE, MEDIAN, AND WIN/TIE/LOSS COUNT.

Evidence	Scope	Cells represented	Semantic	GDOR	Strict	GDOR reduction	Failure/median context	W/T/L
DYN-15	TUM walking_xyz	9	2.296	2.112	2.122	8.0%	—	—
DYN-15	Bonn6 mean	54	12.837	4.079	8.526	68.2%	crowd2 52.184→3.226; med. 4.988→3.577	6/0/0
DYN-15	All7 mean	63	11.331	3.798	7.611	66.5%	crowd2 52.184→3.226; med. 4.278→3.261	7/0/0
DYN-18	TUM walking_xyz	9	2.347	2.119	2.101	9.7%	—	—
DYN-18	TUM walking_halfsphere	9	3.416	3.344	3.350	2.1%	—	—
DYN-18	TUM walking_static	9	0.848	0.843	0.745	0.6%	—	—
DYN-18	TUM sitting_halfsphere	9	32.374	7.354	7.182	77.3%	failure case; paired gain 25.020	3/0/0
DYN-18	TUM4 mean	36	9.746	3.415	3.345	65.0%	sitting 32.374→7.354; med. paired gain 0.209	9/0/3

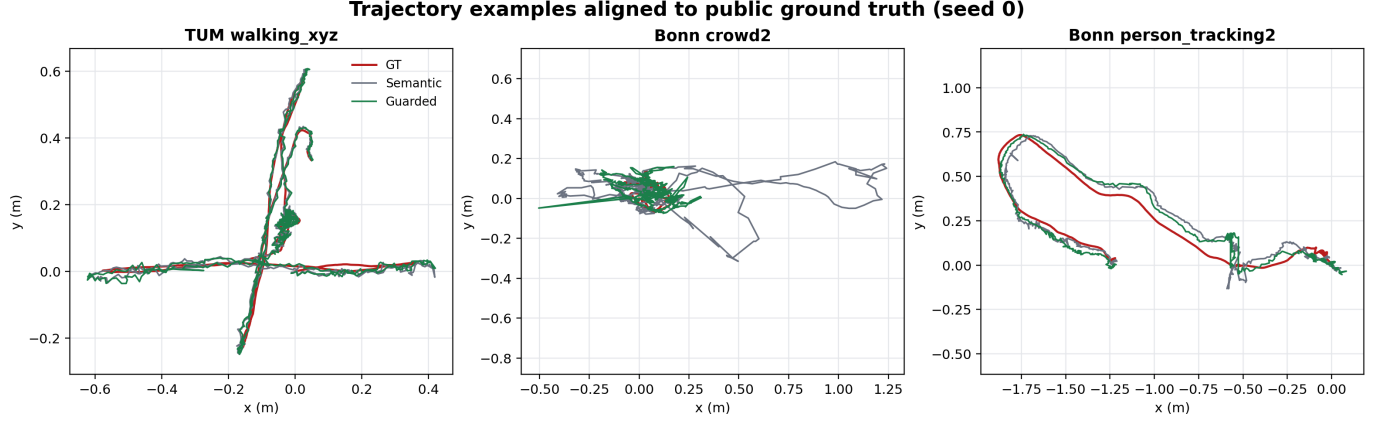


Fig. 3. Seed-0 trajectory examples aligned to public ground truth from the DYN-15 experiment family. The plots are qualitative diagnostics; Tables I and III use all input frames and three-seed means.

TABLE II

DYN-17 MAP-MATCHED NO-TRACK-REUSE CONTROL. MATCHED AND GDOR USE THE SAME PERSISTENT MAPPING WEIGHT AND RAW SEMANTIC-PLUS-FLOW SUPPORT, WHILE ONLY GDOR ENABLES QUALIFIED TRACKING RECOVERY AND ADAPTIVE REPLENISHMENT. ATE IS IN CENTIMETERS. THE MEAN ROW REPORTS FAILURE-CASE, MEDIAN, AND WIN/TIE/LOSS CONTEXT BESIDE THE AVERAGE.

Sequence	Semantic	Matched	GDOR	Map support	GDOR vs. Matched	Median/context	W/T/L
TUM walking_xyz	2.296	2.167	2.112	63.13%	2.5%	—	—
Bonn crowd2	52.184	13.597	3.226	56.73%	76.3%	failure case	1/0/0
Bonn person_tracking2	7.224	7.685	6.821	70.66%	11.2%	—	—
Mean	20.568	7.816	4.053	63.50%	48.1%	crowd2 13.597→3.226; med. gain 0.864	3/0/0

single-seed and covers two scenes. A multi-seed common-manifest mapping study is required for a quantitative mapping-superiority claim. Finally, the experiments establish the combined GDOR mechanism and a matched persistent-weight/no-track-reuse control; they do not claim that any one internal cue independently causes the entire gain.

V. DISCUSSION

A. Why Privilege Separation Matters

Dynamic multimedia capture exposes a persistence asymmetry. A tracking error affects the current pose and may propagate to later states, whereas an unsafe Gaussian write can remain visible from many future viewpoints. Hard exclusion minimizes the second risk but may increase the first when a dynamic mask removes most of the image support. GDOR does not attempt to classify every recovered pixel as truly static. Instead, it asks a narrower question: whether the pixel is

sufficiently consistent with the current camera-induced motion and temporal background to support transient localization. This narrower decision is why recovery can be useful without changing the static-map contract.

The matched control strengthens this interpretation. It shares GDOR’s raw semantic-plus-flow tracking support before recovery and the same persistent mapping weight. Its substantial improvement over Semantic on Bonn crowd2 therefore reflects the combined raw flow mask and persistent-weight route, not a mapping-only intervention. Its remaining gap to GDOR shows that the qualified tracking recovery and adaptive replenishment add value on the tested subset without changing the persistent mapping weight.

B. Regime Dependence and Failure Boundaries

The results do not imply that every masked region should be recovered. On DYN-18 walking_static and walk-

Failure interval, recovered tracking support, and matched control

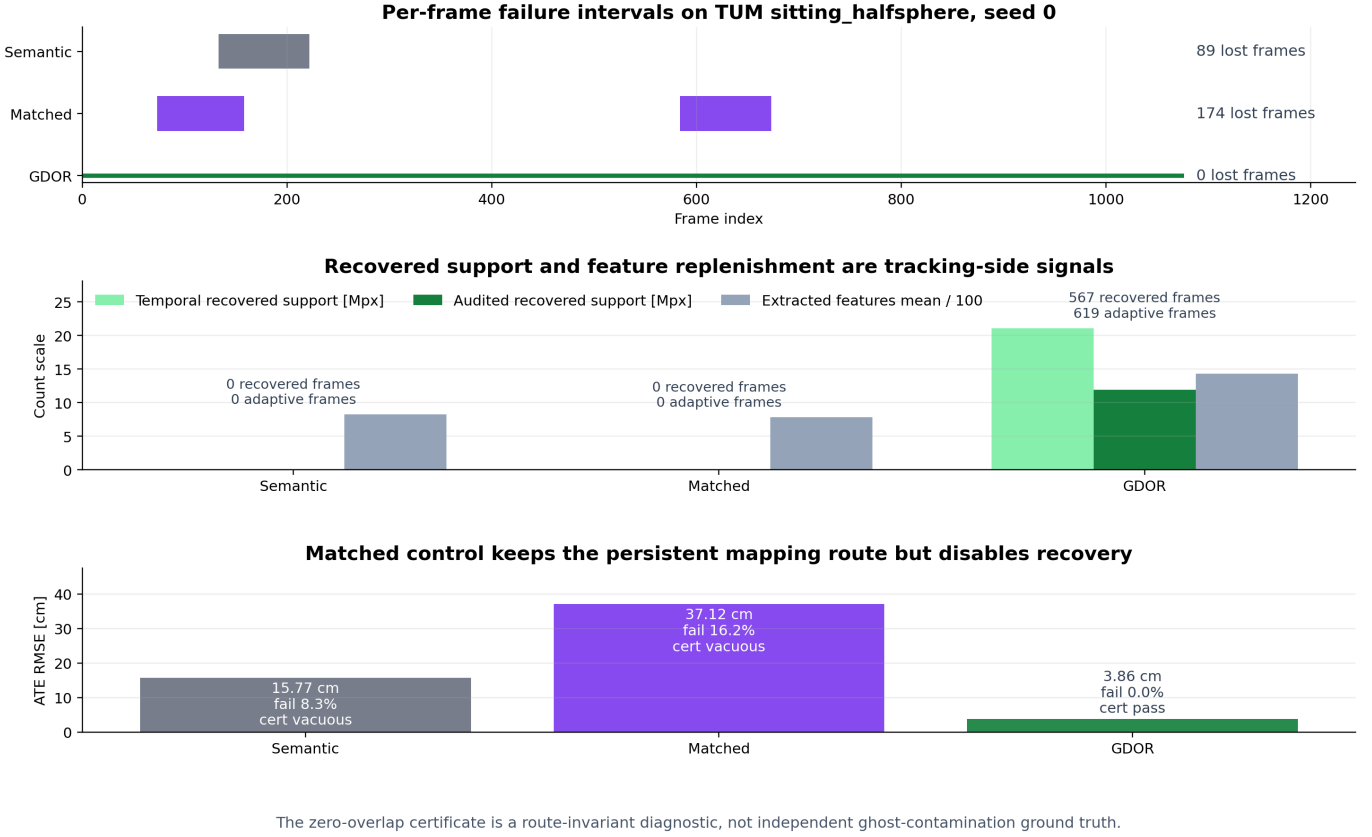


Fig. 4. Representative DYN-19 route diagnostic on TUM sitting_halfsphere, seed 0. Top: failure intervals from per-frame tracking logs. Middle: recovered support and adaptive feature activity are observed only in GDOR; the plotted support pixels are tracking-side evidence, not persistent map admissions. Bottom: the matched control shares the persistent mapping weight but disables qualified tracking recovery. The GDOR certificate for this episode records 567 recovered-support frames, 11.918M audited recovered-support pixels, and zero observed overlap with map-admissible support; this is a route-invariant diagnostic, not an independent ghost-contamination ground truth.

ing_halfsphere, the Semantic parent is already stable and GDOR’s mean improvement is below 3%. The larger gain occurs on sitting_halfsphere, where semantic exclusion removes a large portion of the image and the parent experiences repeated invalid intervals. This pattern is consistent with conditional rescue: recovery is most valuable when exclusion creates an observability deficit.

The Strict control exposes another boundary. It is numerically best on several TUM4 rows, but loses substantially to GDOR on the broader Bonn matrix and costs more runtime. More detailed motion modeling and tighter gates do not produce a monotonic accuracy improvement because prediction quality, intervention coverage, and tracker state interact. The method choice therefore uses the broader DYN-15 accuracy–stability tradeoff rather than selecting the lowest number from the later TUM4 validation.

GDOR also does not eliminate all local errors. Semantic remains competitive on several translational RPE rows, and GDOR is not uniformly best in rotational RPE. The positive claim is sequence-level ATE rescue and reduced failure under the declared protocols, not universal dominance for every local-motion metric.

C. Persistent Map Interpretation

The source routes recovered tracking support and persistent mapping weight independently and implements a direct overlap audit, but the archived DYN-15–18 summaries do not contain that newer counter. The available MapPoint and RGB-D densification counters show that static-weight checks actively reject candidates; they do not empirically certify zero recovered-support overlap for those archived runs. The common-view diagnostic adds an output-level check: under fixed views and fixed static masks, GDOR does not degrade the two evaluated static-region reconstructions and improves the reported mean metrics. However, this evidence is only a limited preservation diagnostic. It is not a complete dynamic-contamination projection over every Gaussian and every viewpoint, and it is not an independent zero-ghost or zero-contamination proof.

A stronger mapping certificate would require multiple seeds, a shared held-out manifest across more scenes, geometric completeness, and an independent ghost-contamination metric. Until that experiment is complete, mapping should be described as preserved under the declared diagnostic rather than generally superior.

TABLE III

SAFETY, EFFICIENCY, AND COMMON-VIEW MAPPING AUDIT. PANEL A REPORTS THE DYN-15 ALL7 MEAN. PANEL B REPORTS DYN-16 SEED-0 ONLINE STATIC-REGION RENDERING OVER TWO SEQUENCES AND 40 SHARED VIEWS. MAPPING VALUES ARE DIAGNOSTIC RATHER THAN MULTI-SEED SUPERIORITY EVIDENCE.

Panel	Method	ATE [cm] ↓	t-RPE [cm] ↓	Failure [%] ↓	Time [s/run] ↓	Map support [%]
A: DYN-15 All7	Semantic	11.331	4.277	0.785	32.46	100.00
	GDOR	3.798	3.945	0.144	49.84	63.31
	Strict	7.611	4.556	0.318	63.61	63.31
Panel	Method	PSNR ↑	SSIM ↑	LPIPS ↓	Views	Seed
B: DYN-16 static region	Semantic	19.068	0.690	0.354	40	0
	GDOR	20.154	0.740	0.287	40	0
	Strict	20.177	0.738	0.287	40	0

D. Reproducibility and External Validity

The clean source release contains the GDOR implementation, configurations, tests, and benchmark scripts but excludes datasets, model weights, maps, trajectories, and experiment outputs. The private experiment ledger retains the authoritative run roots and source hashes. The current release passes the C++ build and test suite and 107 Python tests in the clean publication tree.

External validity and concurrent-method differentiation remain the main blockers to a broader ranking. Published DyPho-SLAM numbers are stronger than the local GDOR values on the four TUM sequences, but the implementations and protocols are not matched. DL-SLAM also overlaps with the high-level tracking–mapping separation premise, so GDOR’s claim is restricted to its RGB-D/ORB routing mechanism and same-source evidence. The correct next experiment is therefore a frozen official-baseline rerun, not threshold adjustment on GDOR. If that runner remains unavailable, the final paper should retain the current same-source claim and present published systems only as related-work context.

VI. CONCLUSION

GDOR-SLAM implements an auditable two-route contract for dynamic-region evidence in an RGB-D/ORB Gaussian SLAM pipeline. Temporal depth, residual flow, and tracking-risk evidence qualify a conservative subset of excluded observations for transient localization, while persistent ORB and Gaussian mapping remain governed by the conservative map mask. This separation allows tracking utility without automatic map-writing authority.

The 63-run TUM/Bonn protocol and separately frozen 36-cell TUM4 validation show consistent improvement over the same-source Semantic parent. The map-matched no-track-reuse control further shows that a fixed raw-mask tracking route with the same persistent mapping weight does not match GDOR on the tested three-sequence subset. The single-seed common-view audit is retained only as a limited preservation diagnostic: it indicates that the tracking intervention does not require degrading the declared static-region Gaussian rendering views, but it does not prove zero ghost contamination or broader mapping superiority.

The current evidence supports the specific guarded recovery and routing mechanism; it does not claim conceptual priority

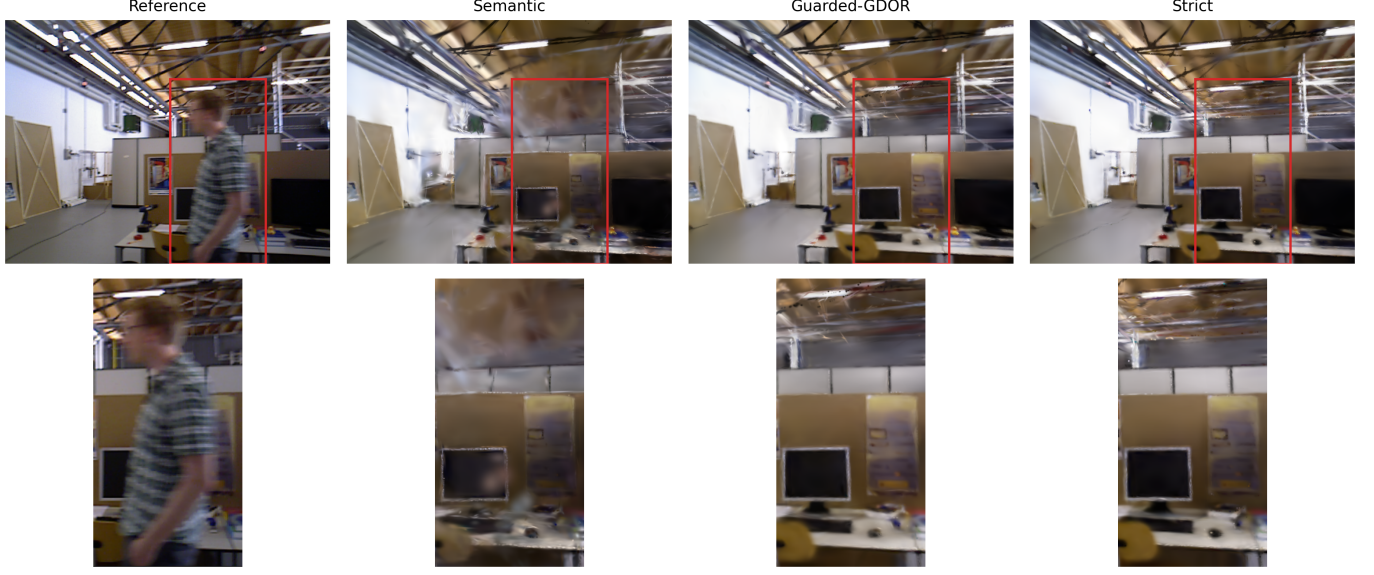
for tracking–mapping separation. It does not yet establish superiority over the official DyPho-SLAM implementation or multi-seed mapping superiority. Closing those two protocol gaps is the required next step for broader external claims.

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] J. Huai, F. Xie, and H. Wang, “Photo-slam: Real-time simultaneous localization and photorealistic mapping for monocular, stereo, and rgb-d cameras,” in *CVPR*, 2024.
- [3] N. Keetha, J. Karhade, K. M. Jatavallabhula, G. Yang, S. Scherer, D. Ramanan, and J. Luiten, “Splat, track & map 3d gaussians for dense rgb-d slam,” in *CVPR*, 2024.
- [4] C. Yan, D. Qu, D. Wang, D. Xu, Z. Wang, B. Zhao, and X. Li, “Gs-slam: Dense visual slam with 3d gaussian splatting,” in *arXiv preprint arXiv:2311.11700*, 2023.
- [5] V. Yugay, Y. Li, T. Gevers, and M. R. Oswald, “Gaussian-slam: Photo-realistic dense slam with gaussian splatting,” in *arXiv preprint arXiv:2312.10070*, 2023.
- [6] B. Bescos, J. M. Fácil, J. Civera, and J. Neira, “Dynaslam: Tracking, mapping, and inpainting in dynamic scenes,” *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 4076–4083, 2018.
- [7] C. Yu, Z. Liu, X.-J. Liu, F. Xie, Y. Yang, Q. Wei, and Q. Fei, “Ds-slam: A semantic visual slam towards dynamic environments,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 1168–1174.
- [8] L. Xiao, J. Wang, X. Qiu, Z. Rong, and X. Zou, “Dynamic-slam: Semantic monocular visual localization and mapping based on deep learning in dynamic environment,” *Robotics and Autonomous Systems*, vol. 117, pp. 1–16, 2019.
- [9] Y. Xu, H. Jiang, Z. Xiao, J. Feng, and L. Zhang, “Dg-slam: Robust dynamic gaussian splatting slam with hybrid pose optimization,” in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.08373>
- [10] M. Kong, J. Lee, S. Lee, and E. Kim, “Dgs-slam: Gaussian splatting slam in dynamic environment,” *arXiv preprint arXiv:2411.10722*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.10722>
- [11] Y. Liu, K. Fan, B. Lan, and H. Liu, “Dypho-slam: Real-time photorealistic slam in dynamic environments,” in *IEEE International Conference on Multimedia and Expo*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.00741>
- [12] L. Zhang, Y.-A. Liu, X. Jiang, C. Huang, D. Li, and Y. Zhang, “Dags-slam: Dynamic-aware 3dgs slam via spatiotemporal motion probability and uncertainty-aware scheduling,” *arXiv preprint arXiv:2602.21644*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.21644>
- [13] R. Mur-Artal, J. Montiel, and J. D. Tardós, “Orb-slam: A versatile and accurate monocular slam system,” *IEEE Transactions on Robotics*, vol. 31, no. 5, pp. 1147–1163, 2015.
- [14] C. Campos, R. Elvira, J. J. Rodríguez, J. M. Montiel, and J. D. Tardós, “Orb-slam3: An accurate open-source library for visual, visual-inertial, and multimap slam,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.

TUM walking_xyz frame 202

Held-out frame 202 at 1341846320.631954908 s



Bonn crowd frame 201

Held-out frame 201 at 1548339826.611599922 s

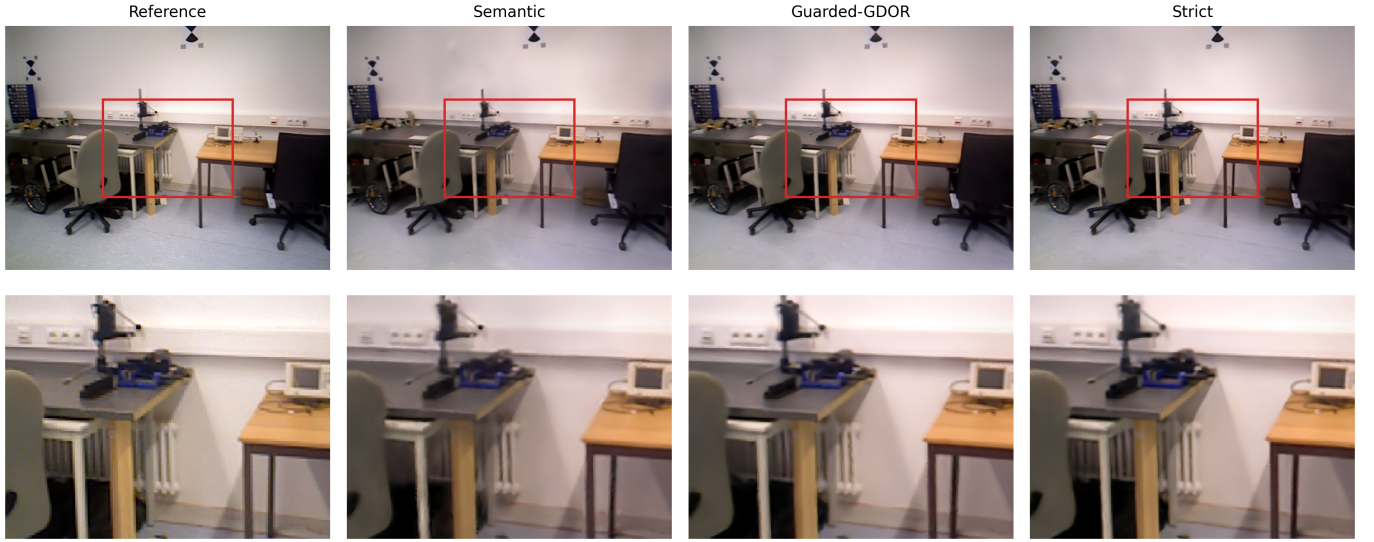


Fig. 5. DYN-16 common-view online mapping diagnostic. Each row uses the same held-out frame, fixed static-region mask, and final-map render for Semantic, GDOR, and Strict. Dynamic people in the reference are excluded from the static-region metric.

- [15] T. Whelan, R. F. Salas-Moreno, B. Glocker, A. J. Davison, and S. Leutenegger, "Elasticfusion: Real-time dense slam and light source estimation," *International Journal of Robotics Research*, vol. 35, no. 14, pp. 1697–1716, 2016.
- [16] W. Tan, H. Liu, Z. Dong, G. Zhang, and H. Bao, "Robust and accurate slam in unstructured environments," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 5459–5466.
- [17] E. Palazzolo, J. Behley, P. Lottes, P. Giguere, and C. Stachniss, "Refusion: 3d reconstruction in dynamic environments for rgb-d cameras exploiting residuals," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 3193–3200.
- [18] J. Zhang, M. Henein, R. Mahony, and V. Ila, "Vdo-slam: A visual dynamic object-aware slam system," in *arXiv preprint arXiv:2005.11052*, 2020.
- [19] B. D. Lucas and T. Kanade, "An iterative image registration technique with an application to stereo vision," in *International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.
- [20] E. Sucar, S. Liu, J. Ortiz, and A. J. Davison, "imap: Implicit mapping and positioning in real-time," in *ICCV*, 2021, pp. 6229–6238.
- [21] Z. Zhu, S. Peng, V. Larsson, W. Xu, H. Bao, Z. Cui, M. R. Oswald, and M. Pollefeys, "Nice-slam: Neural implicit scalable encoding for slam," in *CVPR*, 2022, pp. 12 786–12 796.
- [22] X. Yang, H. Li, H. Zhai, Y. Ming, Y. Liu, and G. Zhang, "Vox-fusion: Dense tracking and mapping with voxel-based neural implicit representation," in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2022, pp. 499–507.
- [23] X. Hu, C. Zhang, M. Zhao, Y. Gui, X. Zhang, and X. Ji, "Dygs-slam: Real-time accurate localization and gaussian reconstruction for dynamic scenes," in *International Conference on Computer Vision*, 2025, pp. 9561–9571. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Hu_DyGS-SLAM_Real-Time_Accurate_Localization_and_Gaussian_Reconstruction_for_Dynamic_Scenes_ICCV_2025_paper.html
- [24] C. Huang, L. Zhang, T. Deng, K. Wang, and M. Li, "Motion probability-based 3dgs-slam in dynamic environment," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025, pp. 14 045–14 052.

- [Online]. Available: <https://ieeexplore.ieee.org/document/11247059>
- [25] Z. Xu, Q. Li, X. Liu, C. Chen, and J. Niu, "DI-slam: Enabling high-fidelity gaussian splatting slam in dynamic environments based on dual-level probability," *arXiv preprint arXiv:2607.01860*, 2026. [Online]. Available: <https://arxiv.org/abs/2607.01860>
- [26] S. Zhu, Y. Huang, W. Wu, C. Jiang, Y. Chen, I.-M. Chen, and H. Wang, "D²gsam: 4d dynamic gaussian splatting slam," *arXiv preprint arXiv:2512.09411*, 2025. [Online]. Available: <https://arxiv.org/abs/2512.09411>
- [27] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of rgb-d slam systems," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 573–580.